EXP NO:18 Experiment to understand KNN algorithm for a given data set

AIM:
To understand the KNN algorithm for the data set given.

PROGRAM:

python

```python
import numpy as np
import pandas as pd

# Read the dataset
df = pd.read_csv("/content/student_admission.csv")

# Display dataset information
df.info()

# Count admission categories
df.Admitted.value_counts()

# Display first 5 rows
df.head()

# Check missing values
print(df.isnull().sum())

# Rows with no label can't be used for training, so drop them
df = df.dropna(subset=['Admitted'])

# Fill missing feature values
df['Age'] = df['Age'].fillna(round(df['Age'].median()))
df['EntranceScore'] = df['EntranceScore'].fillna(round(df['EntranceScore'].mean()))

# Reset index
df.reset_index(drop=True, inplace=True)

# Confirm no missing values remain
print(df[['Age', 'EntranceScore', 'Admitted']].isnull().sum())

# Select features and label
features = df.iloc[:, [1, 2]].values
label = df.iloc[:, 3].values

# Import libraries
from sklearn.model_selection import train_test_split
from sklearn.neighbors import KNeighborsClassifier

# Split the dataset
xtrain, xtest, ytrain, ytest = train_test_split(
    features, label, test_size=0.2, random_state=42
)

# Create and train KNN model
model_KNN = KNeighborsClassifier(n_neighbors=5)
model_KNN.fit(xtrain, ytrain)

# Check accuracy
print(model_KNN.score(xtrain, ytrain))
print(model_KNN.score(xtest, ytest))

# Confusion matrix
from sklearn.metrics import confusion_matrix
print(confusion_matrix(label, model_KNN.predict(features)))

# Classification report
from sklearn.metrics import classification_report
print(classification_report(label, model_KNN.predict(features)))
```

OUTPUT:

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 10 entries, 0 to 9
Data columns (total 4 columns):
 #   Column         Non-Null Count  Dtype
---  ------         --------------  -----
 0   Department     9 non-null      object
 1   Age            9 non-null      float64
 2   EntranceScore  9 non-null      float64
 3   Admitted       10 non-null     object
dtypes: float64(2), object(2)
memory usage: 452.0+ bytes
Department       1
Age              1
EntranceScore    1
Admitted         0
dtype: int64
Age              0
EntranceScore    0
Admitted         0
dtype: int64
0.75
0.5
[[0 3]
 [0 7]]
              precision    recall  f1-score   support

          No       0.00      0.00      0.00         3
         Yes       0.70      1.00      0.82         7

    accuracy                           0.70        10
   macro avg       0.35      0.50      0.41        10
weighted avg       0.49      0.70      0.58        10
```

RESULT:

Thus the program to understand the KNN algorithm for the given data set was executed successfully.